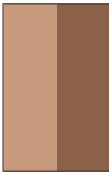
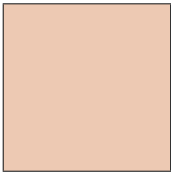
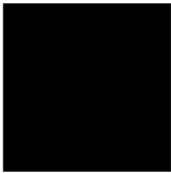
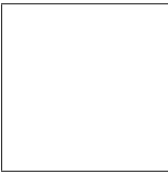


The app walks this card in about 30 seconds and writes `device_calib.json` = {white-balance gains, px-per-mm, focus-ok}. **Macro and face numbers do not count without a calibration under 7 days old** (EVAL.md §7). If a patch reads grey instead of the value printed under it, the white balance is wrong — re-run A before anything else.

A

Gray patches — white balance and exposure



white (target 1.00)

18% gray (0.82)

black (0.00)

skin, light

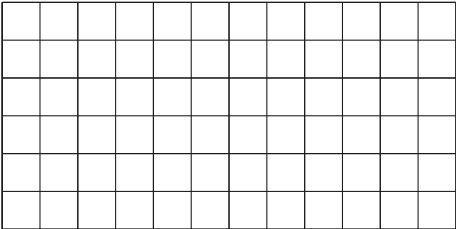
med / deep

22 mm squares
(2 mm patch + margin)
is the working FOV

A1 seat the clip on the white patch and capture. A2 capture the 18 % gray. A3 capture the skin patch — this is the one that catches the orange-peel shift a cheap clip adds, and it is what the macro classifier is normalised against. A4 confirm the black patch is not crushed to 0 at both capture extremes; if it is, the driver is saturating, which is a hardware fault and not a threshold to tune.

B

The 5 mm grid — is the cell square?



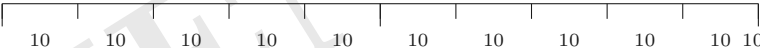
60 mm × 30 mm, 5 mm grid, 10 mm lines heavy

Count cells across a 2 mm macro patch. Fewer than 0.4 cells means the patch is below the noise floor and the capture is GREY — not RED, never RED (FUSION.md).

Un-square cells here mean a bad pixel pitch or a badly seated clip, long before the classifier notices anything.

C

The 1 mm scale — pixels per millimetre

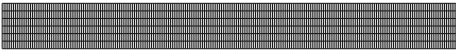
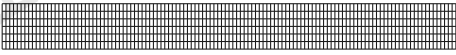
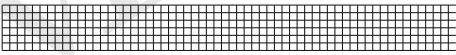


1 mm alternating bars, 100 mm long — the px-per-mm ruler

Count pixels across ten alternating bars. Divide by ten. That number is `pxPerMm` in `device_calib.json` and it is what makes a 2 mm patch comparable across devices and clips. Two devices agreeing to within 2% is the repeatability check in step 6. 10 mm ruler — for the manifest and for your notes, not for the app

D

Resolution wedges — how sharp is sharp enough



1.0 mm pitch (6 px at 6 px/mm) — offset halftone typically lands here

0.5 mm pitch (3 px) — inkjet halftone typically lands here

0.25 mm pitch (1.5 px) — starts to close up

0.125 mm pitch (0.75 px) — below sensor Nyquist: expect mush, and that is correct

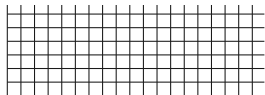
Report the *coarsest* group whose bars still separate. Six millimetre bar height.
That report is the sharpness bar the app shows during capture (SPEC FR-C3).
_____ bar-height wedge, 0.5 mm pitch, 1 μm steps: the height at which bars blur is your depth of field

D is a hardware check, not a knob. If the top group already blurs at the seated distance, the clip is not seated or the card is curved — fix the mechanical problem. Never lower the sharpness gate to make a capture pass (AGENTS.md §5: weakening a gate to make a test pass is a review fail).

E

Rosette vs droplet — what R-PROC actually reads

rosette — offset / litho / laser



Regular lattice at a fixed screen angle (15°, 45°, 75°); the macro FFT finds a radial peak at it.
schematic, not to scale

droplets — inkjet



Irregular size and placement; ink spreads into the fibre and the lattice edge softens. The regular peak is gone.
schematic, not to scale

What the model sees:

a radial FFT peak (screen-angle regularity) plus an LBP texture histogram. Both are cheap, and both are sensitive to optics — which is exactly why the clip exists and why this card must be calibrated before any threshold is chosen (BUILD.md §7: never eyeball-tune).

This card does not prove any of it.

It gives the operator a visual reference and gives the model one patch whose substrate is known. Any accuracy or robustness number comes from the eval harness with a run id — never from this page (AGENTS.md §5).

Dye-sub PVC differs again: often little surface relief and a different base fluorescence. That is the UV question, and UV is auxiliary (QA_BANK) — it never carries a RED on its own.

F The 30-second routine (app-guided)

- 1 Clean the clip window. At this magnification dust is a 0.2 mm feature and it will happily masquerade as microprint.
- 2 Seat the shroud flat on section A and wait for the sharpness bar to go green. Section D is the reference for what green means.
- 3 Walk A: white → 18 % gray → skin → black. The app solves white balance and exposure from the first three.
- 4 Walk B for cell shape, then C for px-per-mm.
- 5 Walk D. Record the coarsest wedge group that separates — a number, written down, with the clip id next to it.
- 6 Re-seat the clip three times and re-capture C each time. If px-per-mm moves by more than 2%, the shroud is not repeatable and the clip goes back to the bench.
- 7 Store `device_calib.json`, stamp the calib id in the box at the top right, and write the date. Every macro patch you collect afterwards carries that calib id (DATA.md §3).

G What this card is not

Not a security feature, not a tamper-evident print, not a legal document, and not a genuine identity record — it is a **measurement target**. It carries nobody's personal data. It is watermarked SPECIMEN because a calibration card that looks official eventually ends up stuck to somebody's ID by mistake, and because that call belongs to a government form, not to us.

Bring two laminated copies to the finals (DEMO.md §4). Venue lighting is uncontrolled (RISKS R10). If a macro capture greys out on stage, re-run this routine against the card before anybody touches a threshold.

KASOTI — SIH prototype for evaluation. Not for operational deployment without MHA legal and technical clearance (THREAT_MODEL.md §8). Source: `hardware/calibration_card.tex`, rev A. SPECIMEN — KASOTI TEST